UnderCode

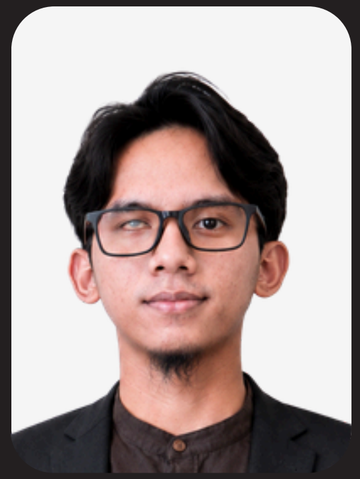
Rakamin Academy

Hire Smarter Not Harder

A data-driven approach to optimizing recruitment efficiency and predicting offer acceptance

GitHub Dataset Python Code Dashboard

Meet the Founding Team



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We're hiring slower, paying more, and losing candidates

What's changing

- Hybrid work
- Global talent pools
- Skill-based hiring

What it causes

More complex sourcing and candidate evaluation

Key challenges

- Longer time-to-hire
- Rising cost-per-hire
- Declining offer acceptance

Source: LinkedIn

What does our data tell us?

Avg. time-to-hire

47.2 days

11.19 days above benchmark
Benchmark: LinkedIn

Avg. cost-per-hire

$5,215

$465 above benchmark
Benchmark: SHRM

Avg. offer acceptance

65%

15% below benchmark
Benchmark: KPI Depot

The right fit, the right yes

Transforming recruitment from reactive and intuition-driven to data-driven — by predicting which candidates are most likely to accept an offer before hiring decisions are made.

Goal

Develop a decision-support system that predicts offer acceptance and enables more efficient, cost-effective hiring.

Objectives

1. Diagnose recruitment inefficiencies using historical hiring data
2. Identify key drivers of offer acceptance
3. Build a binary classification model to predict acceptance and support decision-making

Success Metrics

≥80% AUC-ROC and F1-score for offer acceptance prediction

Model performance threshold

≥80% Target offer acceptance rate

Up from current 65%

Dataset overview

	num_applicants	time_to_hire_days	cost_per_hire	offer_acceptance_rate
count	5,000	5,000	5,000	5,000
mean	155.62	47.19	5,214.83	0.651
std	84.16	23.87	2,731.00	0.202
min	10	7	507.16	0.300
25%	83	26	2,820.60	0.480
50%	157	47	5,218.29	0.630
75%	229	67	7,611.41	0.830
max	299	89	9,998.91	1.000

Data types

Identifier

- recruitment_id

Categorical

- department
- job_title
- source

Numerical

- num_applicants
- time_to_hire_days
- cost_per_hire
- offer_acceptance_rate

Data quality

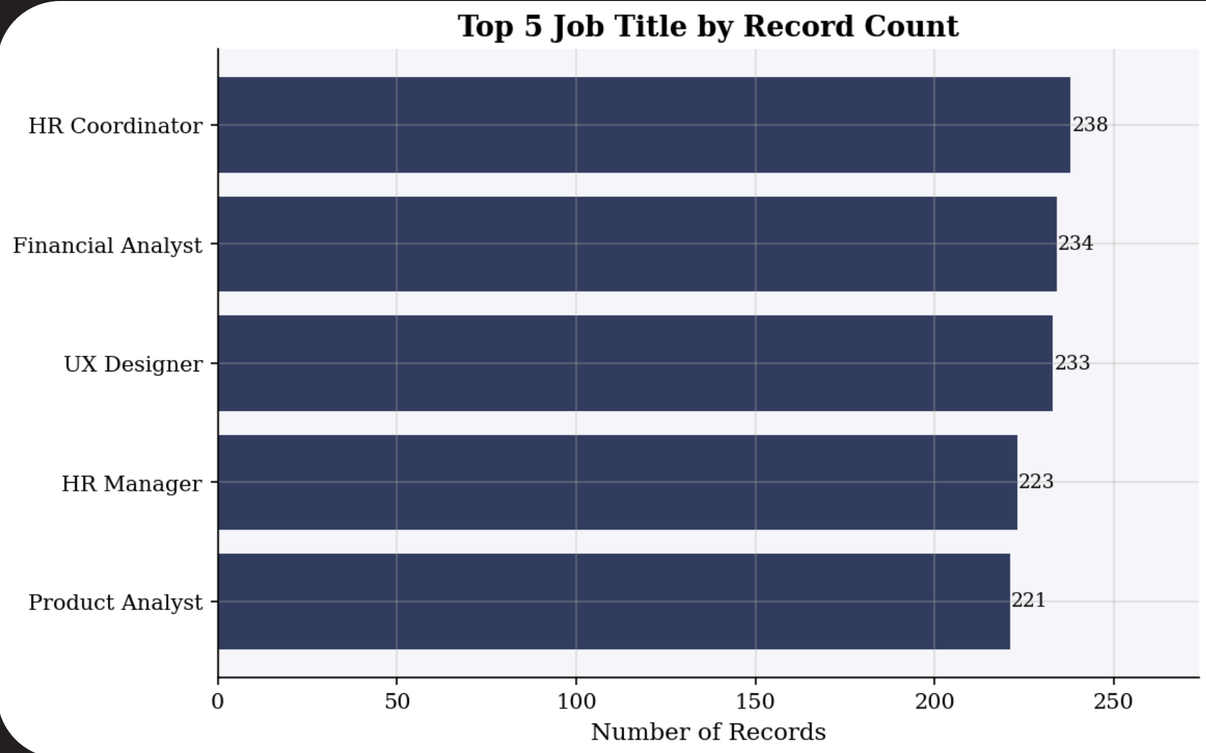
0 missing values
duplicates
outliers

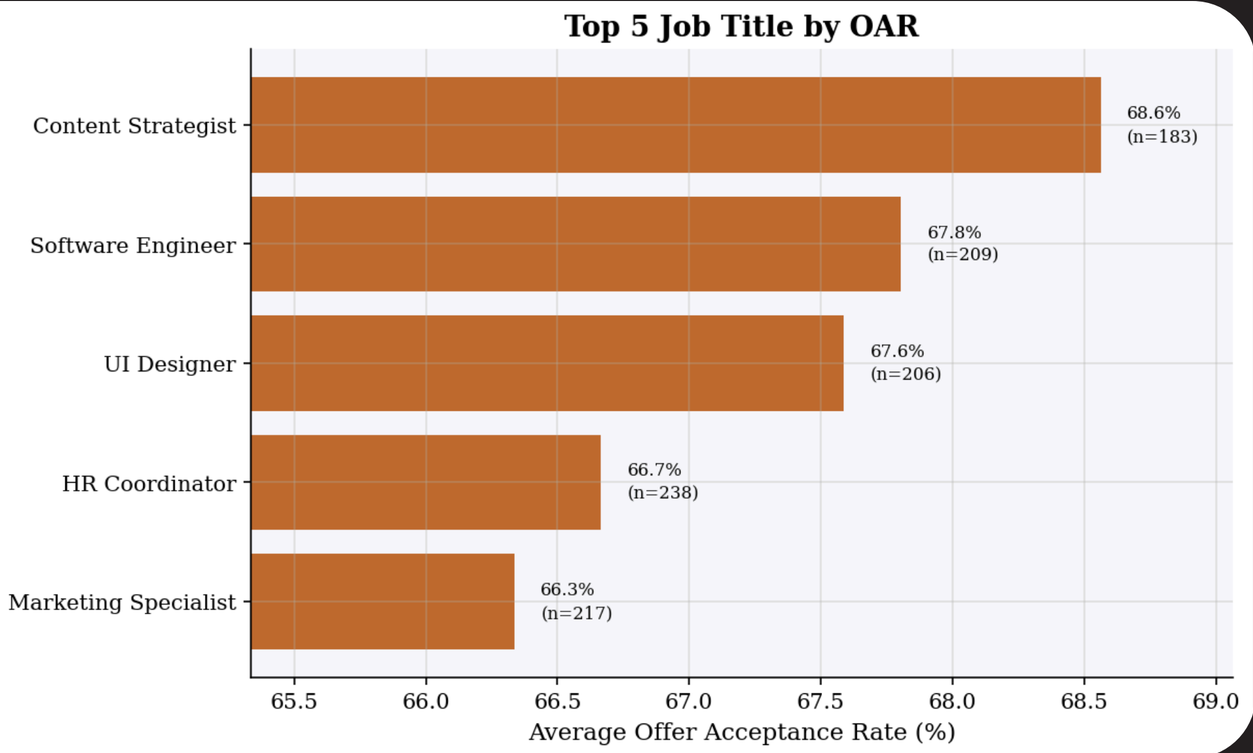
Target feature — Offer Acceptance Rate

- Low OAR (0) — acceptance rate < 70%
- High OAR (0) — acceptance rate ≥ 70%

Threshold based on KPI Depot benchmark

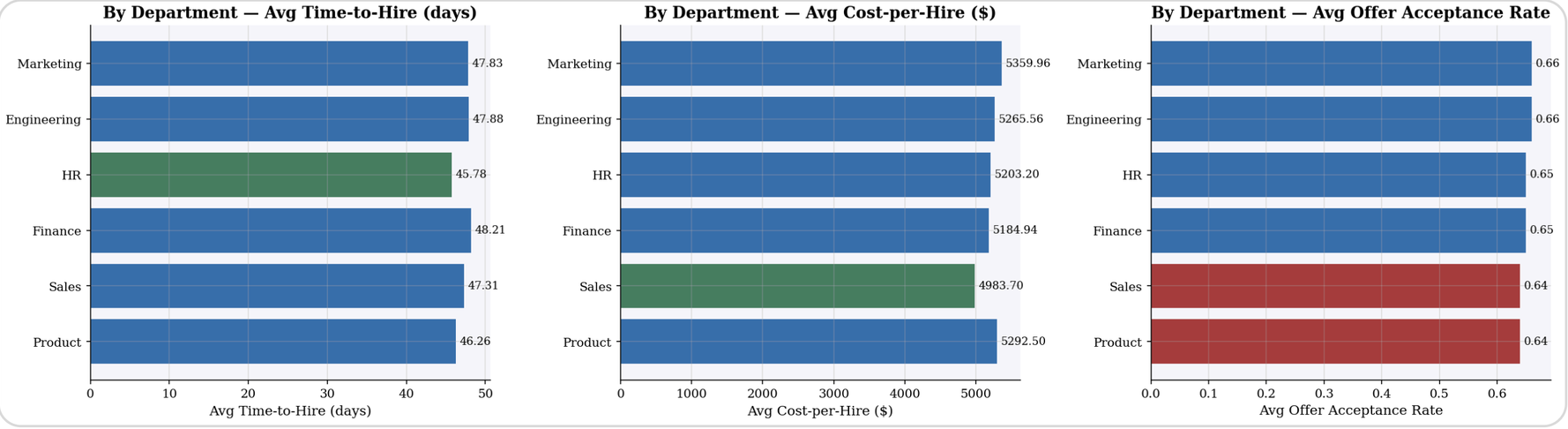
More applicants, not better outcomes





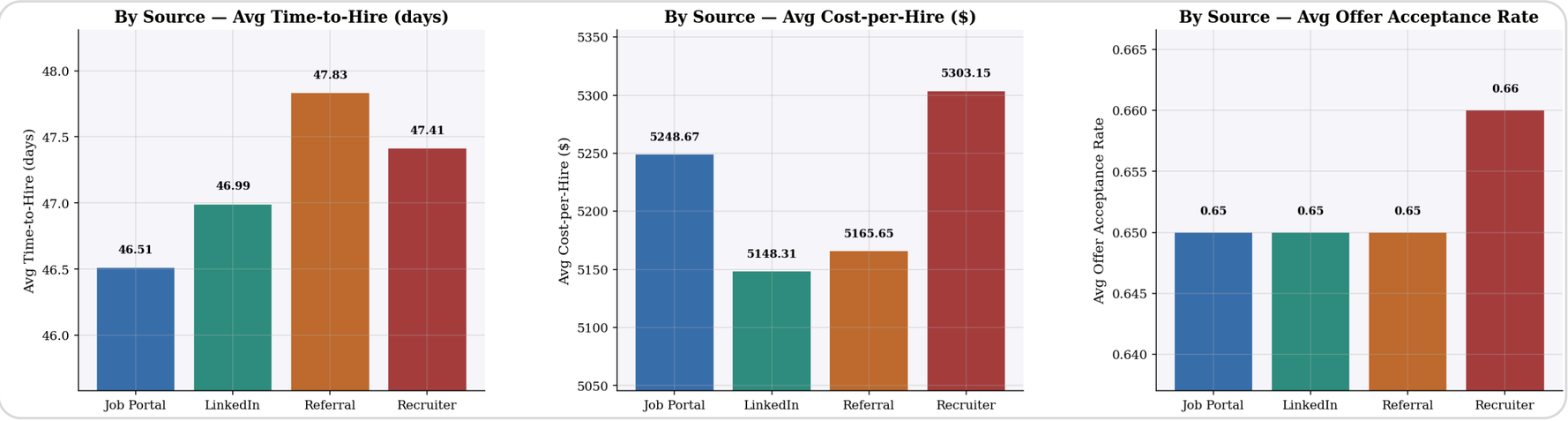
A higher number of applicants does not lead to a higher Offer Acceptance Rate. Roles with large candidate volumes — such as HR Coordinator (238 records) — achieve only moderate acceptance rates (66.7%), while roles like Content Strategist (183 records) top the OAR chart at 68.6%. This indicates that **increasing applicant quantity alone creates additional recruitment cost and longer hiring cycles without improving outcomes.**

Where you hire matters more than who hires



Departments perform equally

All departments show nearly identical TTH, CPH, and OAR — suggesting inefficiencies are **systemic, not department-specific.**

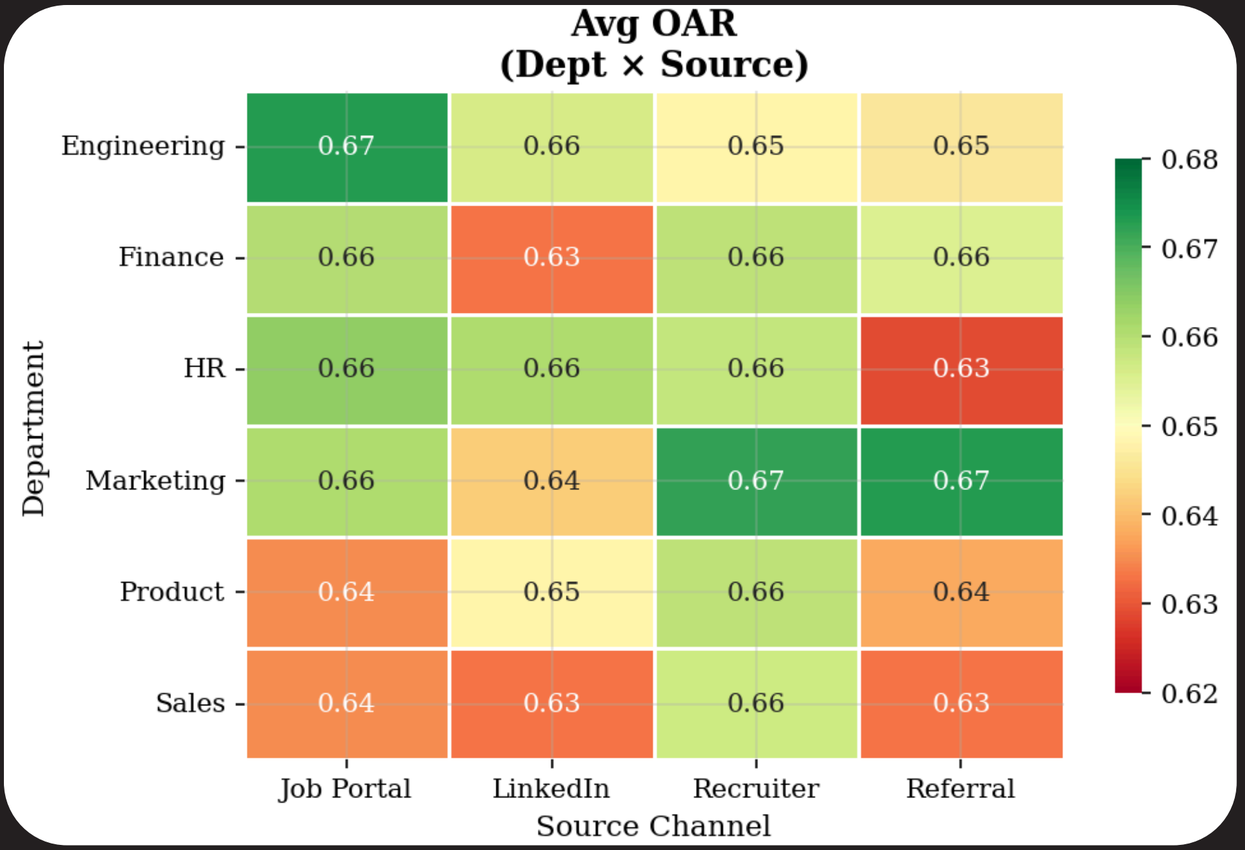


Source channel trade-off

Each channel shows a different efficiency trade-off. **No single channel wins on all three metrics.**

Job Portal — fastest (46.5 days)
LinkedIn — lowest cost ($5,148)
Recruiter — highest OAR (66%)

Match the channel to the role



No single source works best for all roles

OAR varies by department and source channel. **Aligning the right channel to each department** is key to improving acceptance rates.

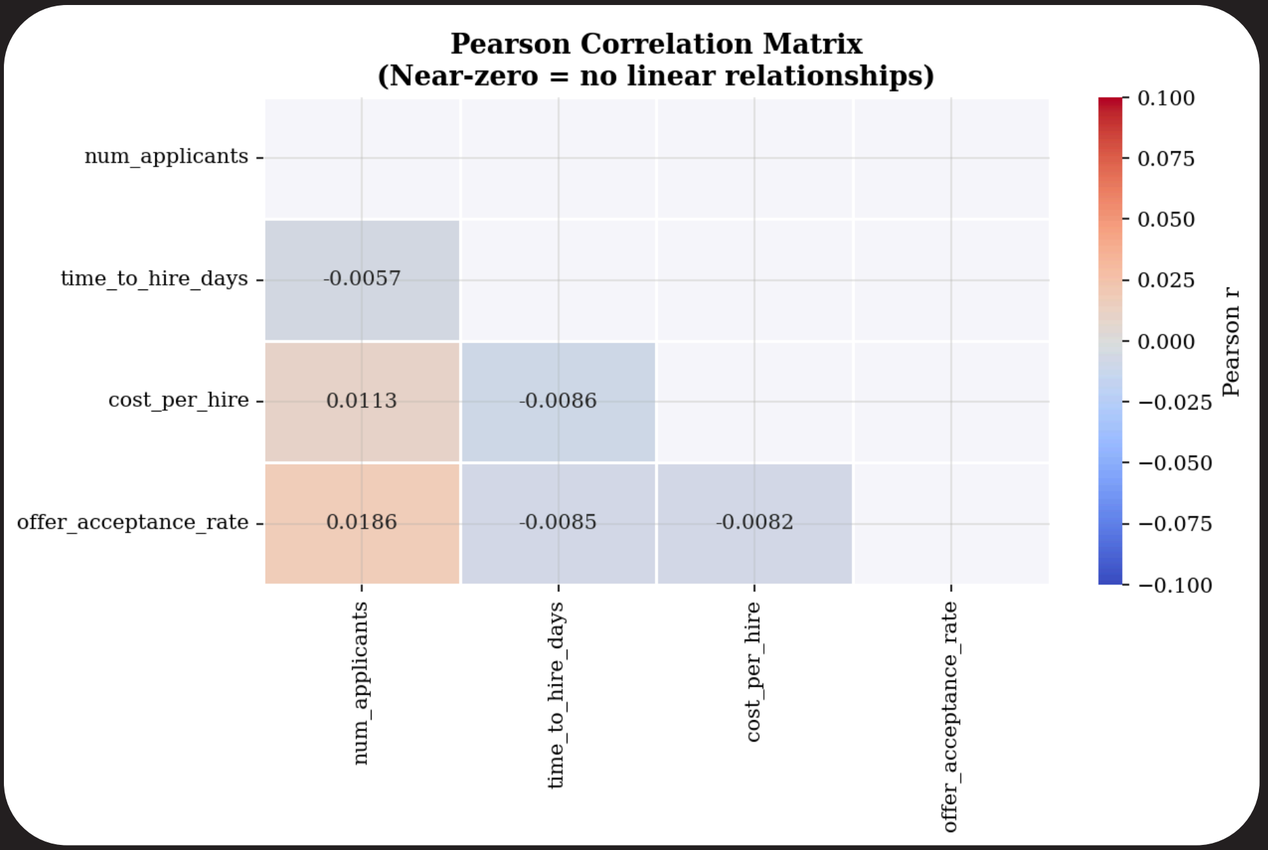
Best channel per department

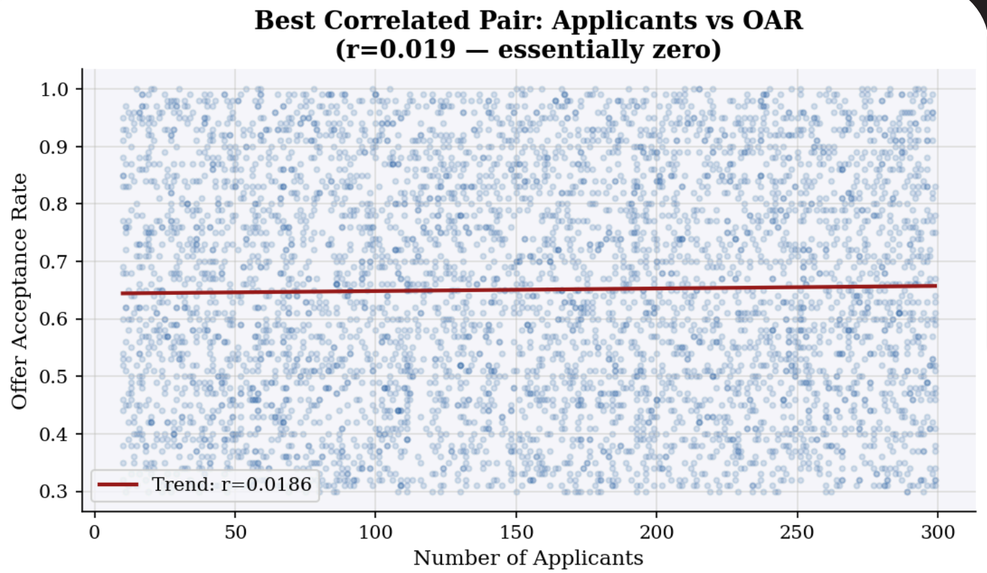
Engineering	Job Portal (0.67)
Finance	Recruiter / Referral (0.66)
HR	Job Portal / LinkedIn / Recruiter (0.66)
Marketing	Recruiter / Referral (0.67)
Product	Recruiter (0.66)
Sales	Recruiter (0.66)

Key takeaway

Recruitment channel effectiveness differs across departments. **Channels that work in Engineering may not perform equally in Product or Sales** — indicating the need for a targeted, department-specific sourcing strategy to improve OAR while reducing unnecessary cost and hiring time.

Numbers don't lie — but they don't always correlate





Increasing candidate volume does not improve OAR
Even the strongest correlation in this dataset (r = 0.019) is **statistically negligible** — the trend line is nearly flat.

No strong linear drivers of offer acceptance identified

Recruitment performance is influenced more by **process quality, candidate fit, and sourcing effectiveness** than by simply increasing hiring activity or candidate volume. This confirms the need for a **predictive, data-driven model** rather than relying on volume-based strategies.

Turning weak signals into strong predictor

Efficiency ratio

- **`cost_per_applicant`**
  → cost_per_hire / num_applicants
- **`cost_per_day`**
  → cost_per_hire / time_to_hire
- **`applicants_per_day`**
  → num_applicants / time_to_hire_days

Role context

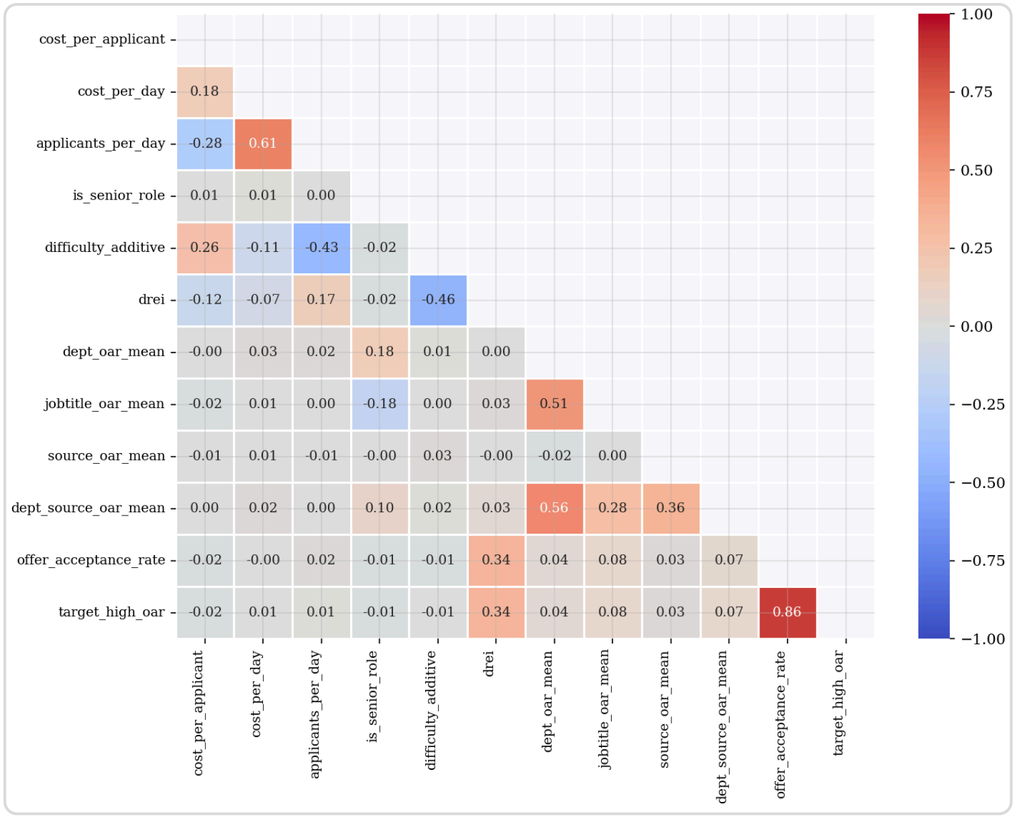
- **`is_senior_role`**
  → keyword flag (strategist, manager, specialist, executive)

Difficulty additive

- **`difficulty_additive`**
  → (zscore(cost_per_hire) + zscore(time_to_hire)) / 2

Target encoding

- **`dept_oar_mean`**
- **`jobtitle_oar_mean`**
- **`source_oar_mean`**
- **`dept_source_oar_mean`**



Engineered features reveal better signals

Raw features showed near-zero correlations. After engineering, t**arget-encoded features (dept, job title, source, dept×source OAR means) capture meaningful patterns** — providing the model with stronger predictive signals than raw inputs alone.

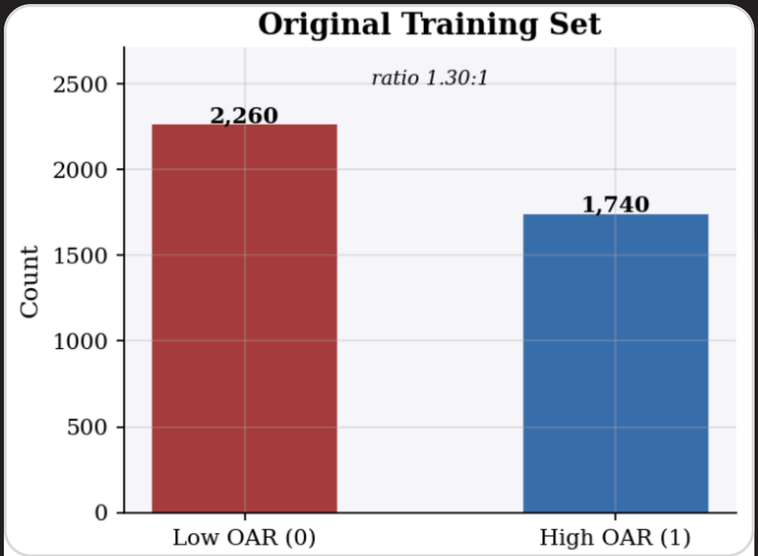
Preprocessing: scaling & balancing the data

Standard scaling

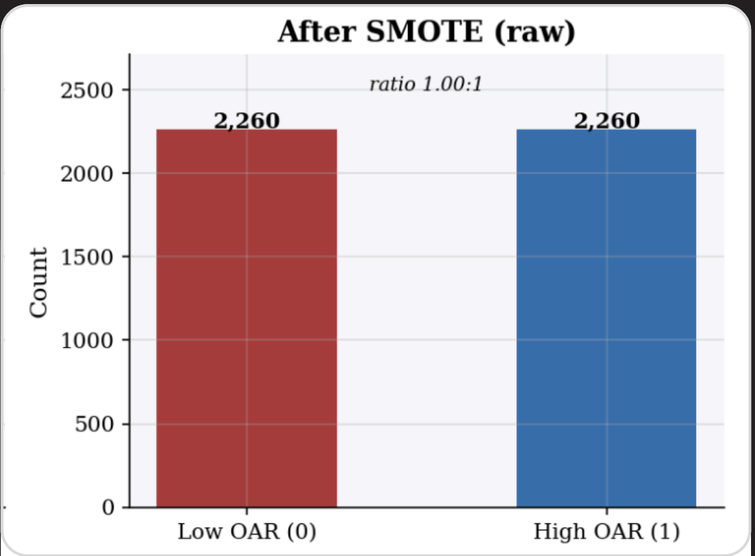
Applied for distance and gradient-based models — **LR, KNN, SVM** — to ensure features are on a comparable scale and improve training stability. Raw (unscaled) data preserved for **tree-based models (XGBoost)** which are insensitive to scaling.

SMOTE — Synthetic Minority Oversampling

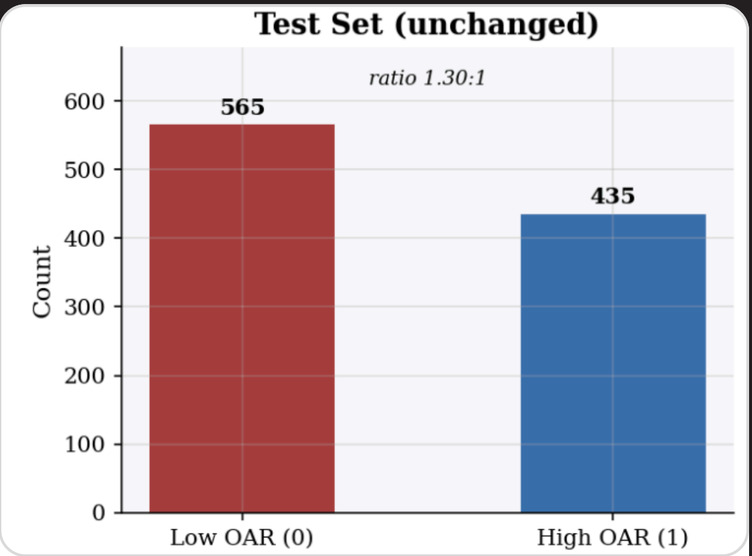
Applied **exclusively to the training set** to address class imbalance by generating synthetic samples for the minority class. Test set kept **unchanged** to maintain fair model evaluation.



Imbalanced



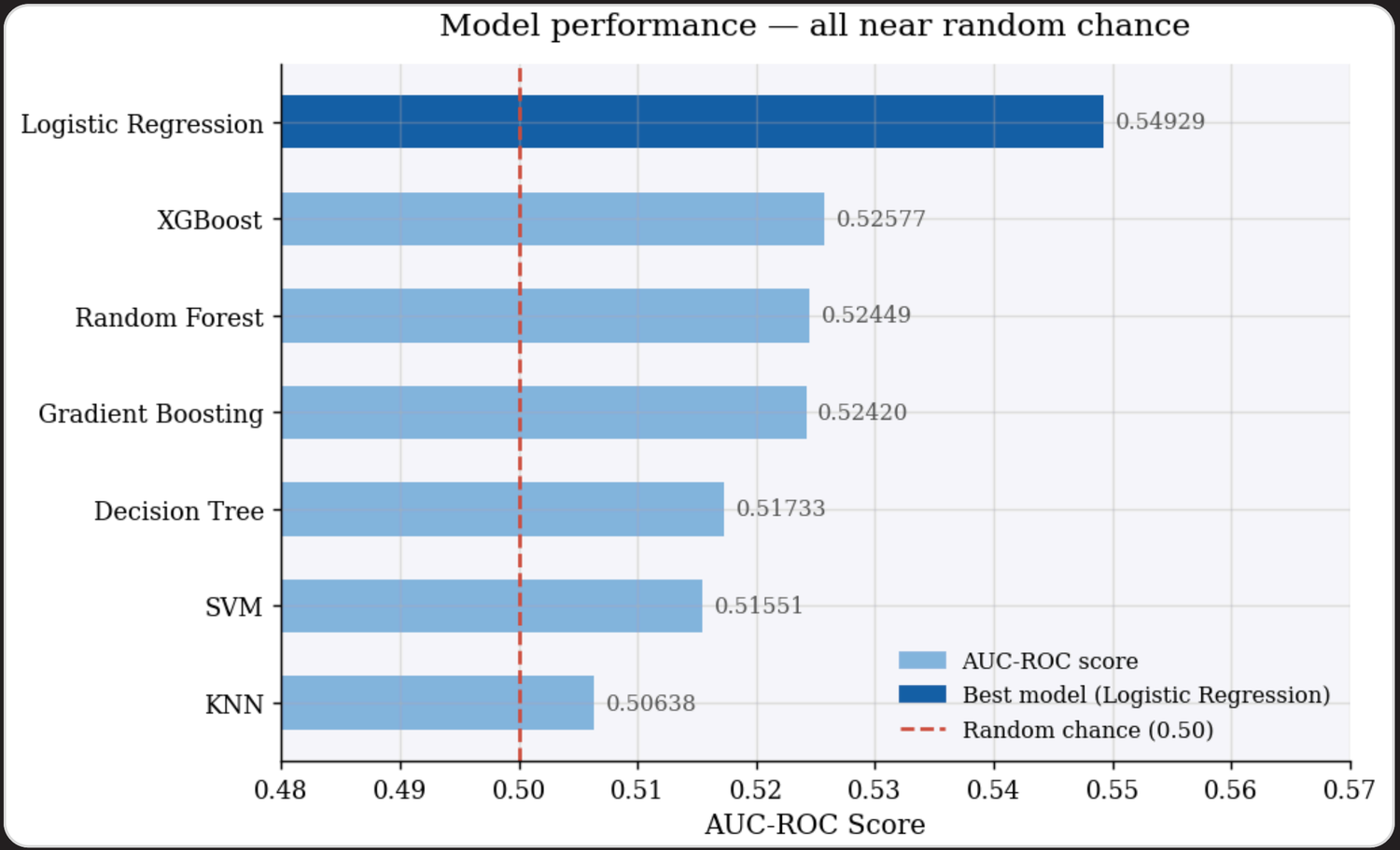
Balanced



Fair evaluation

Our initial models are guessing — not learning

Every model scored close to 0.5 — **no better than a coin flip.** The data we gave them lacks a critical signal: context within each department.



Why are models failing?

All scores cluster between **0.506 – 0.549**, barely above random. The models lack **department-level context** — they can't distinguish a high performer from a low one within the same team.

New Feature: DREI

Department-Relative Efficiency Indicator — a yes/no flag: was this record outstanding compared to its own department?

- ✓ Lower cost than department median
- ✓ Faster hire than department median
- ✓ Higher OAR than department median

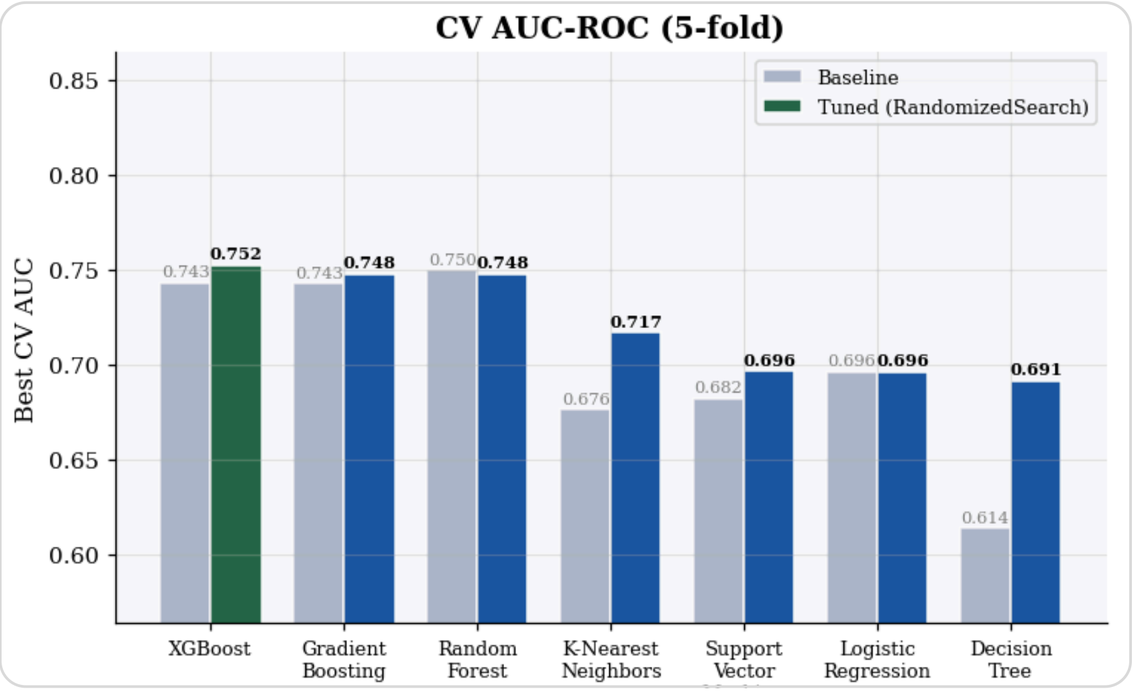
DREI = 1 when all three conditions are met.

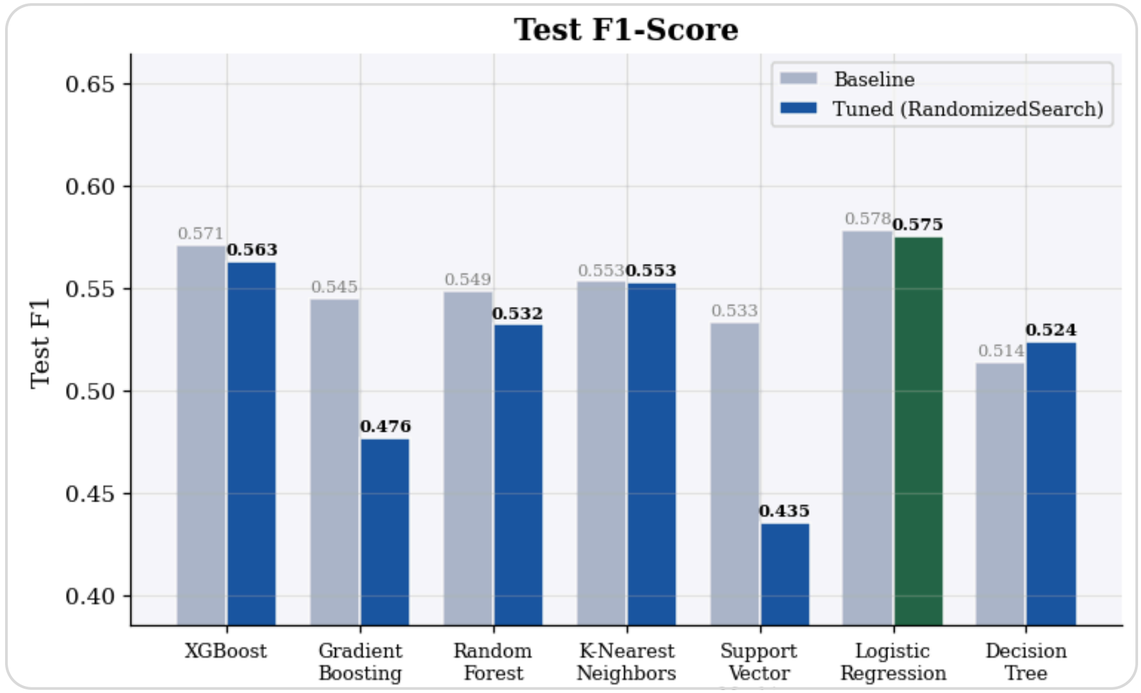
Which model should we deploy?

Comparing all seven models before and after tuning — **AUC** measures risk ranking ability, **F1** measures prediction accuracy.

1. **AUC:** XGBoost leads at **0.752** after tuning, with Gradient Boosting and Random Forest close behind at 0.748. Margins are tight at the top.

2. **F1:** Logistic Regression takes the lead at **0.575**. Three models dropped after tuning (GB, RF, SVM) — a warning sign of **overfitting**.

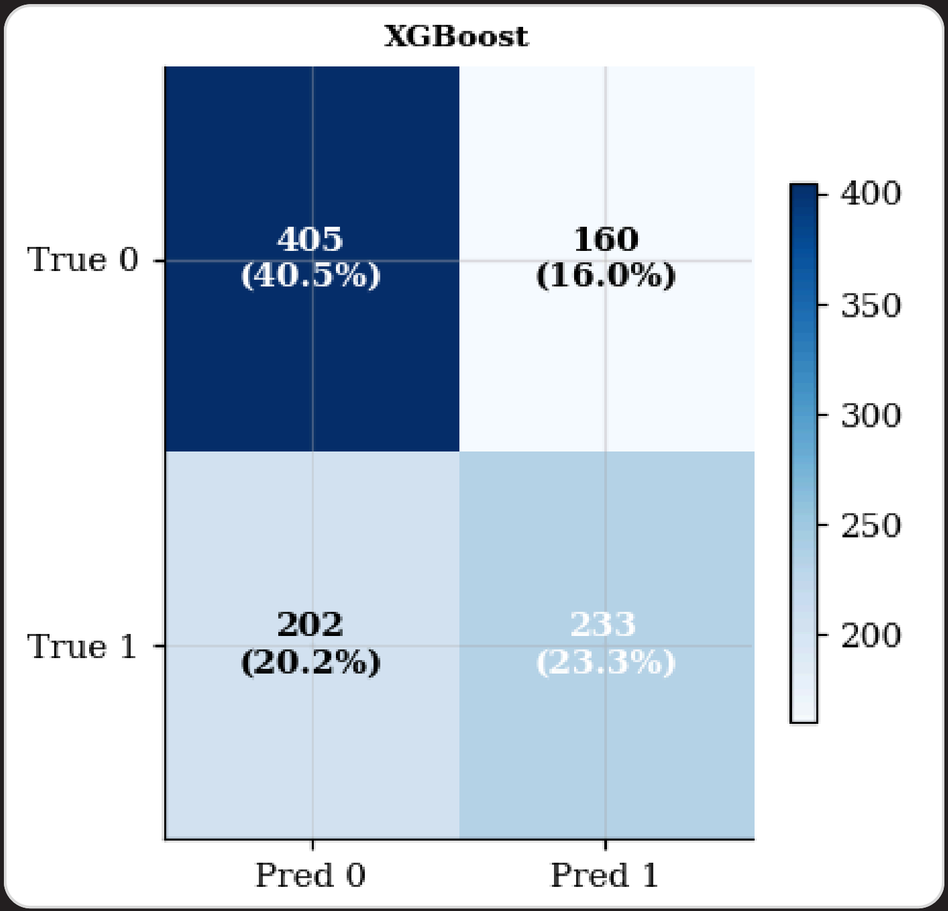




No single model wins on every metric — which is why we evaluate holistically. **XGBoost offers the strongest overall balance** between AUC-ROC and F1, making it the recommended model for deployment.

What the model actually tells HR

Diving into the confusion matrix reveals why XGBoost is our recommended model — it achieves the best trade-off between catching true acceptances and minimizing false alarms.



TN — 405

Recruiter time & cost saved

These candidates were correctly flagged as likely to decline. **HR can deprioritize them early**, saving interview slots, offer prep, and negotiation effort.

TP — 233

Offers worth fast-tracking

Correctly predicted acceptances. **HR can prioritize these candidates**, allocate more attention, and close faster — directly improving OAR.

FP — 160

Wasted hiring effort

Offers extended to candidates expected to accept — but they declined. Each FP = **recruiter time + interview costs + delayed hiring cycle.**

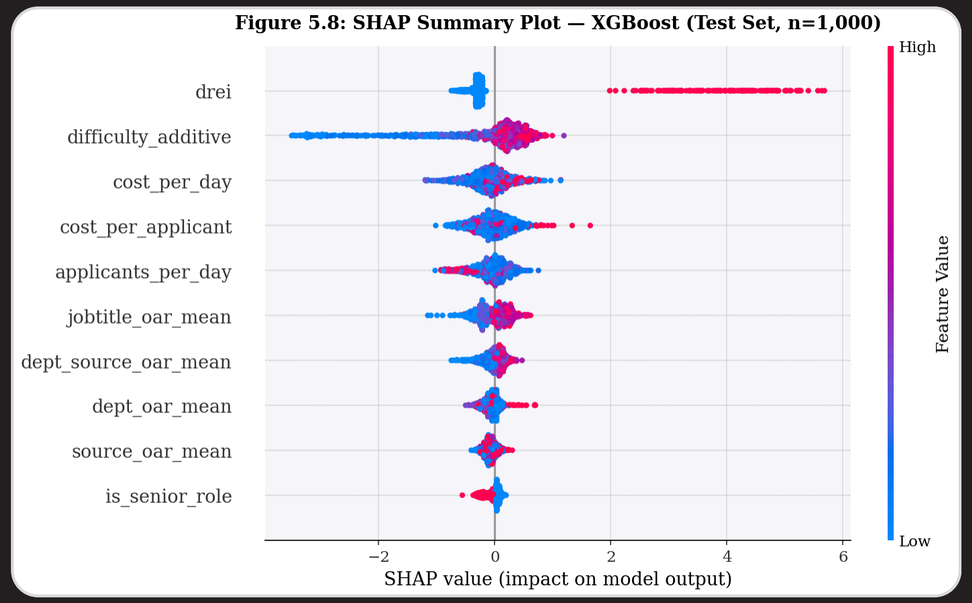
FN — 202

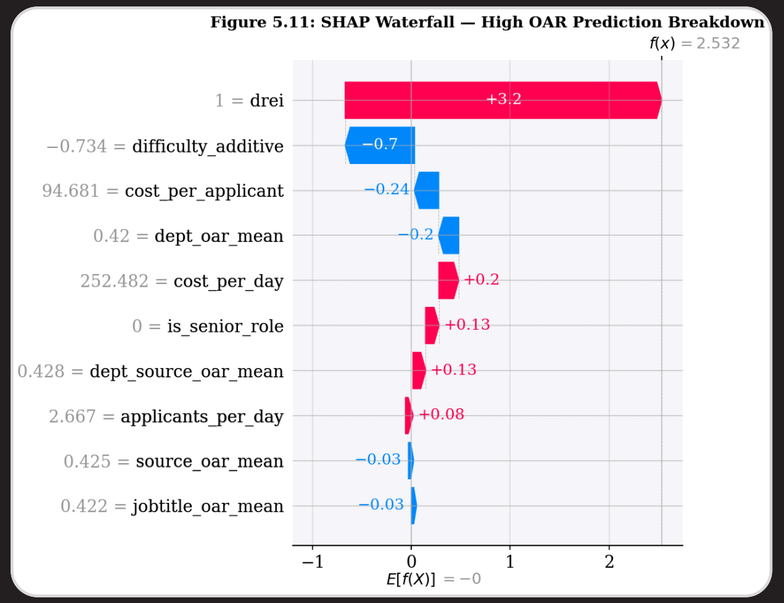
Missed good hires

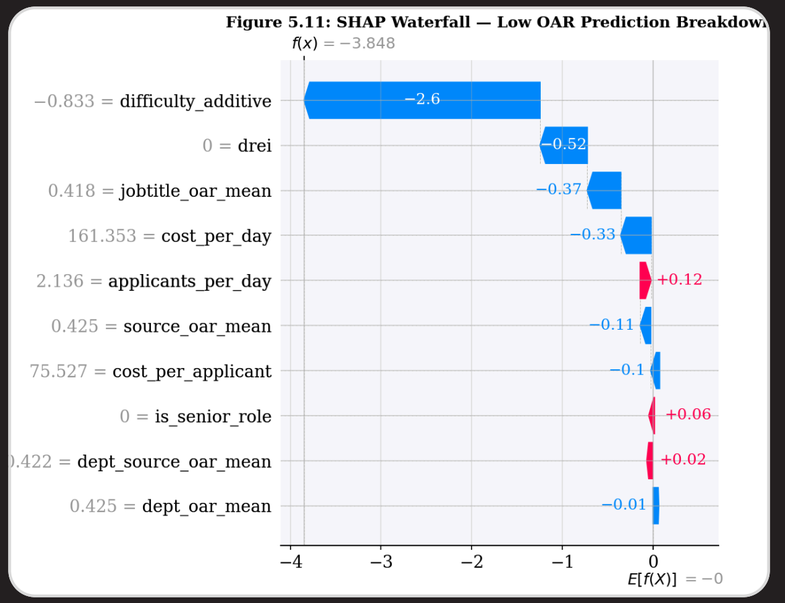
Candidates who would have accepted but were deprioritized. **These are lost hires** — often the costliest error in recruitment.

Why the model predicts what it predicts

SHAP analysis explains XGBoost's predictions — confirming it's explainable, trustworthy, and aligned with recruitment logic.







DREI is the #1 driver

When a candidate outperforms their department on cost, speed, and quality — **the model is highly confident they will accept**. HR should prioritize outreach, reduce delays, and allocate senior recruiter attention.

High difficulty = High rejection risk

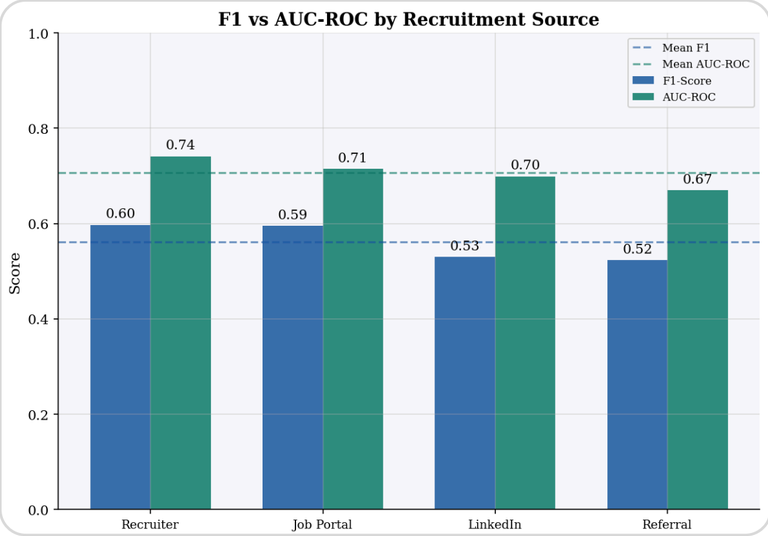
Roles that are both expensive and slow to fill signal a **poor candidate-role fit or sourcing mismatch**. HR should review the job description, switch channels, or adjust compensation before extending offers.

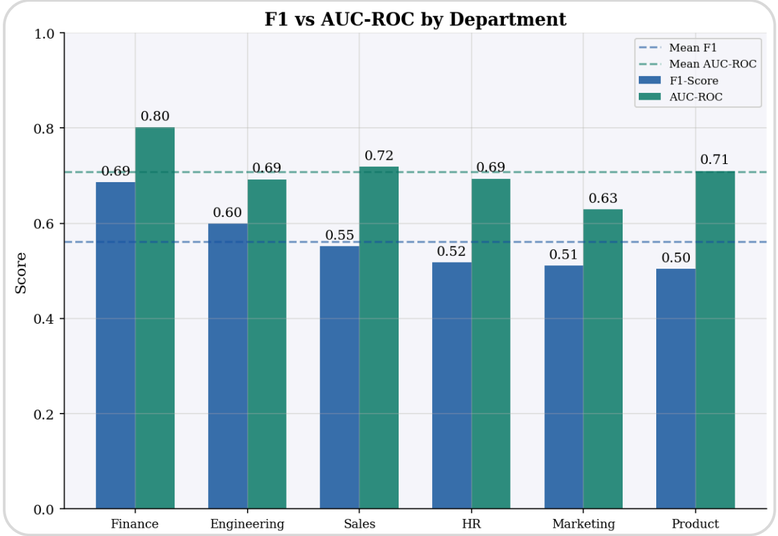
Cost efficiency signals candidate quality

High cost-per-day and cost-per-applicant **negatively impact the prediction**. Efficient hiring processes — fewer applicants, faster closes — correlate with candidates more likely to accept.

Is the model fair across all groups?

A fair model should perform consistently across subgroups. We tested XGBoost across recruitment sources and departments to identify where predictions may be biased or unreliable.





Finance is the model's sweet spot

Highest AUC (0.80) + strong F1 (0.69)
Finance candidates are the most predictable. **HR can trust model predictions most confidently here** — prioritizing Finance pipeline management based on model scores will yield the best ROI.

AUC is consistent

Model ranks fairly across all groups
AUC-ROC stays between **0.67–0.80** across all sources and departments — meaning the model's **ranking ability is reliable** regardless of which group a candidate comes from.

F1 varies — bias risk

Product, Marketing, HR, Referral, & LinkedIn underperform
F1 drops to 0.50 (Product) and 0.52 (Referral) — near random. HR relying on predictions for these groups may make more wrong decisions, creating unfair candidate treatment.

Fairness verdict

The model performs well overall but shows **uneven F1 across departments and sources**. For groups like Product, Marketing, and Referral hires, HR should **combine model scores with human judgment** rather than relying on predictions alone — reducing the risk of systematically disadvantaging certain candidate groups.

Beyond the model — a dashboard HR can use today

The project doesn't stop at finding the best model. We built a ready-to-use recruitment decision dashboard powered by Streamlit, integrated with GitHub for easy access and deployment.

Streamlit

Frontend — Interactive Dashboard

Open-source Python framework for building data apps instantly. No web development experience needed — HR teams can interact with predictions through a clean, browser-based UI.

GitHub

Version Control & Deployment

The dashboard is hosted directly via GitHub integration. Any model update is pushed to the repo and reflected in the live app — **no manual deployment needed.**

XGBoost Model

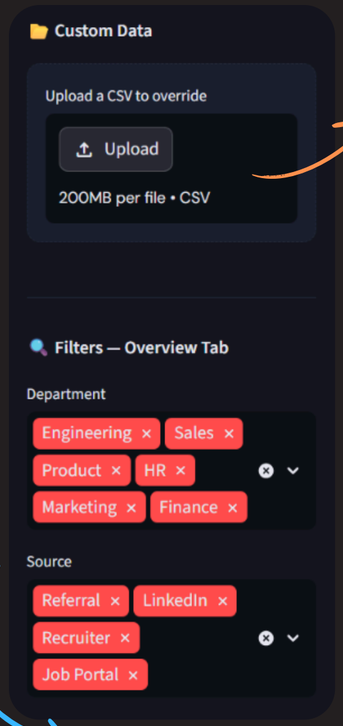
Backend — Prediction Engine

The trained model runs in the background. HR inputs candidate data, and the model returns an **acceptance probability score in real time.**

1. **Candidate Input Form**
   HR enters candidate details — department, source, cost, time-to-hire — and gets an instant **offer acceptance probability score.**
2. **High / Low OAR Classification**
   The model classifies each candidate as **High OAR (≥70%)** or **Low OAR (<70%)** with a confidence level, helping HR prioritize who to pursue.
3. **Recruiter Action Recommendation**
   Based on the score, the dashboard suggests a clear next step — fast-track, review, or deprioritize — turning model output into recruiter actions.
4. **Always Up-to-Date via GitHub**
   Model updates, bug fixes, and new features are pushed to GitHub and **automatically reflected in the live dashboard** — zero manual redeployment.

RePort — Candidate Overview Tab

The first tab gives HR teams a complete, interactive view of their recruitment pipeline — filterable by department and source in real time.



Custom data upload

Upload your own CSV (up to 200MB) to **override the default dataset** — the entire dashboard updates to reflect your organization's actual hiring data.

5 KPI cards

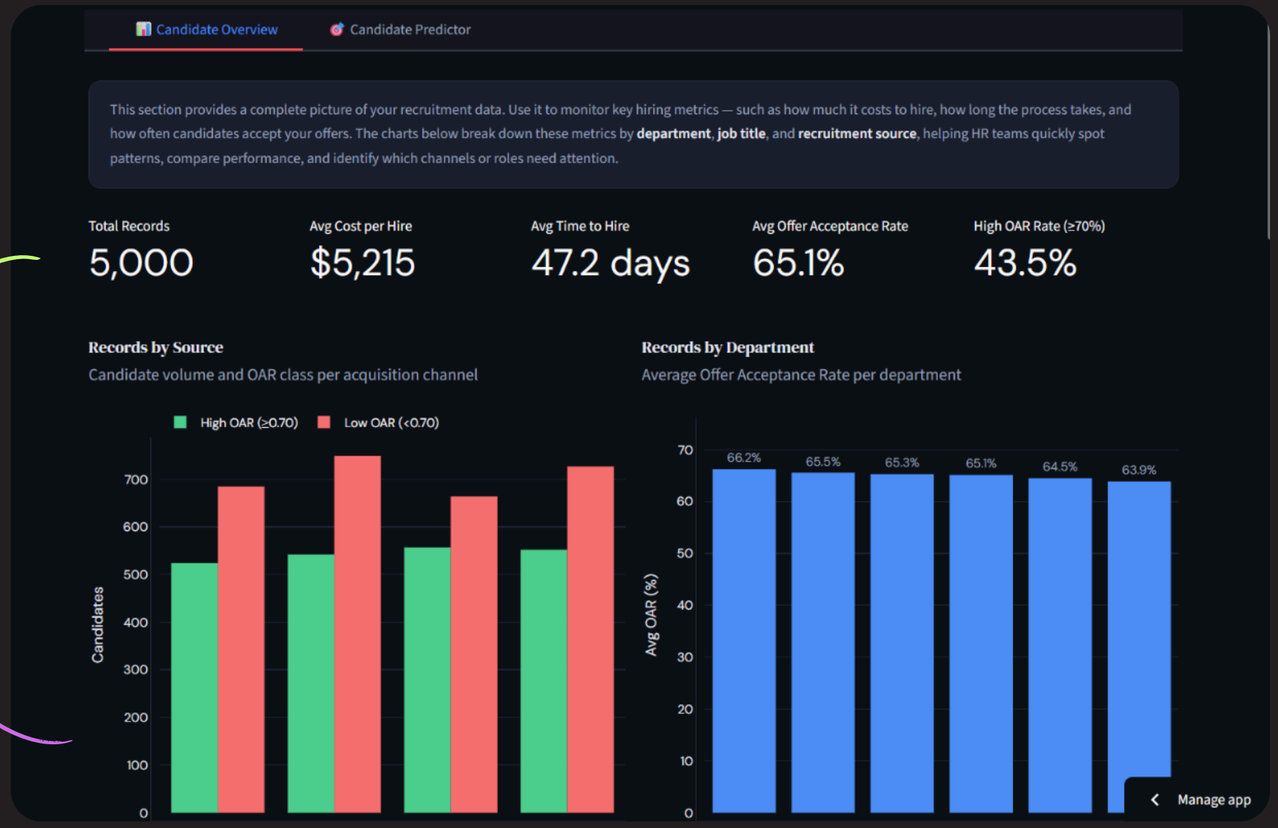
Real-time metrics: total records, avg cost, time to hire, OAR, and High OAR rate — all update when filters change.

Interactive filters

HR can filter by **any combination of department and source** — all charts update instantly, making it easy to drill into specific teams or channels.

Source & Dept charts

Side-by-side High vs Low OAR by source channel, and avg OAR by department — spot which teams and channels underperform.

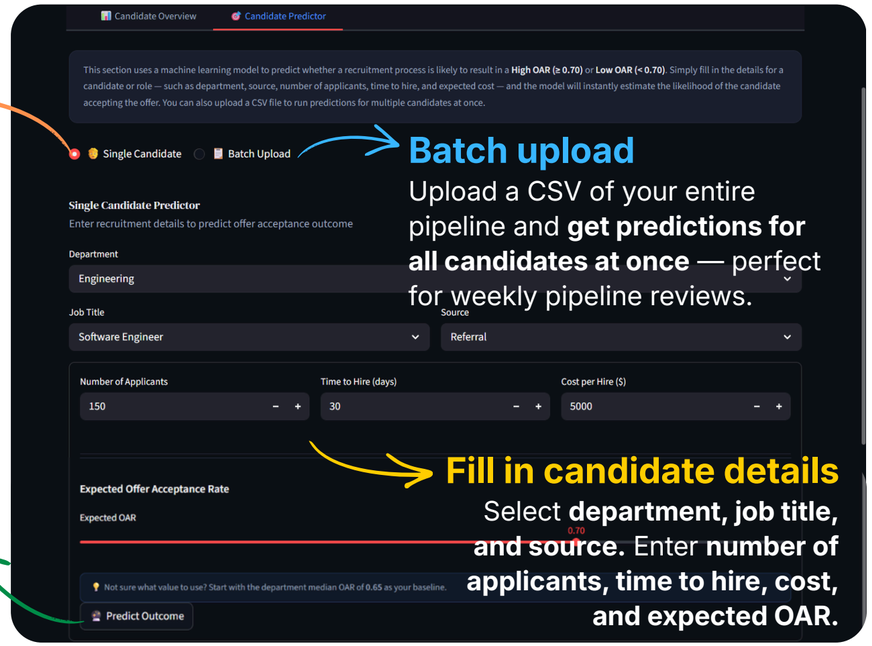


This is just a preview. The full Candidate Overview tab is **interactive, filterable, and ready for your own data**. Open it now and explore every corner of your hiring pipeline. undercode.streamlit.app

RePort — Candidate Predictor Tab

Fill in candidate details and get an instant offer acceptance prediction — single candidate or entire batch. The model explains every score so HR knows exactly what to do next.

Single predictor
Input one candidate at a time with **full control over every parameter** — ideal for live decisions during interviews.

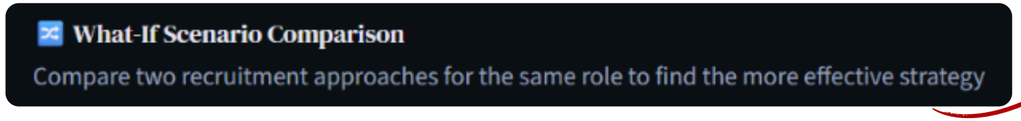


Batch upload
Upload a CSV of your entire pipeline and **get predictions for all candidates at once** — perfect for weekly pipeline reviews.

Fill in candidate details
Select **department, job title, and source.** Enter **number of applicants, time to hire, cost, and expected OAR.**

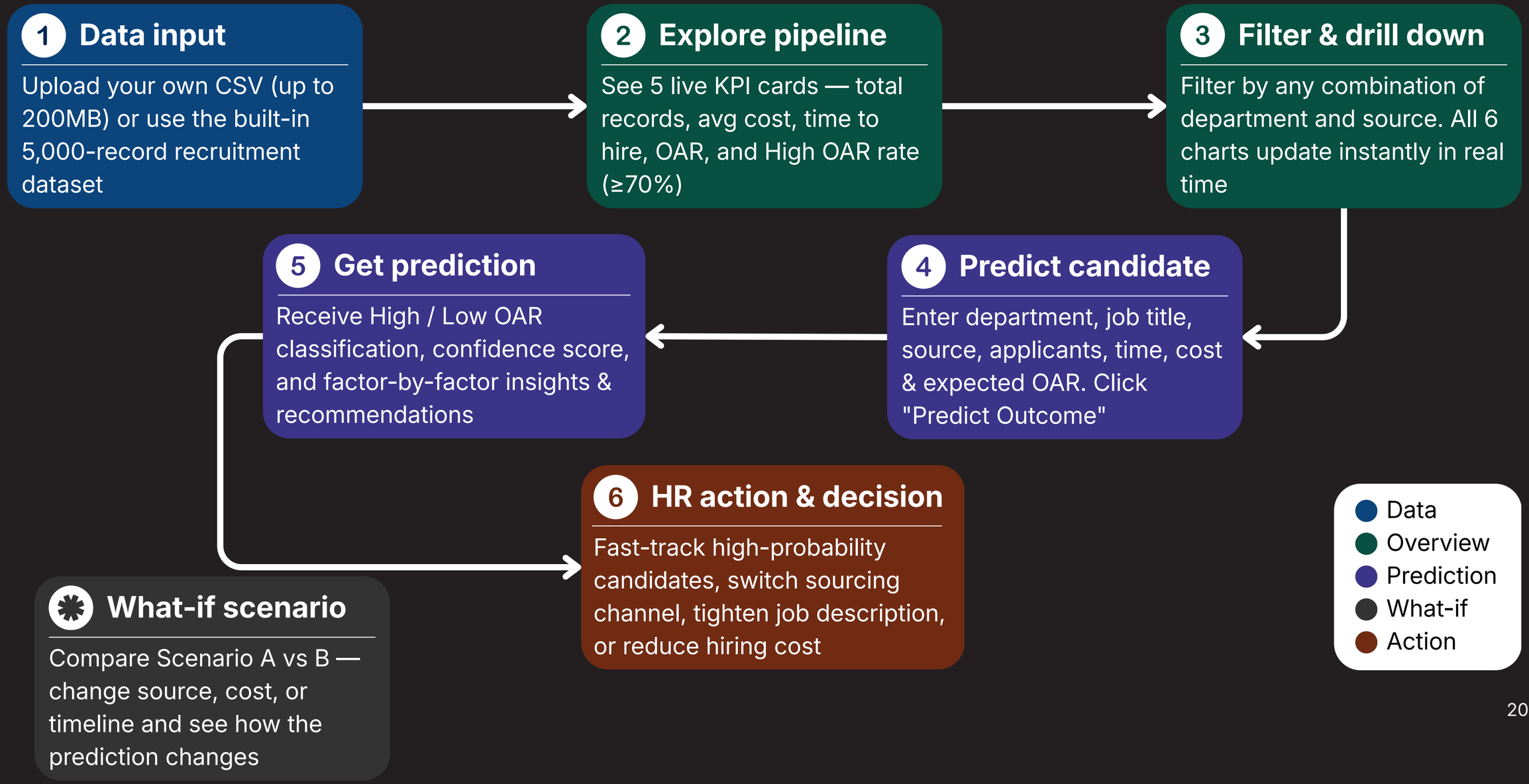
Click "Predict Outcome"
XGBoost runs instantly in the background — **no waiting, no loading screen.**

Get a prediction + explanation
The model doesn't stop at 'yes' or 'no.' It shows you **the confidence behind every prediction** and **breaks down exactly which recruitment factors made the difference** — so your next move is always clear.



"What if you changed the source from Referral to Job Portal? What if you reduced the cost by $2,000? RePort answers these questions instantly — open it now and run your own what-if in under 30 seconds." **undercode.streamlit.app**

RePort — dashboard usage flow



Post-deployment — monitoring & maintenance

Deployment is not the finish line. RePort follows a structured monitoring and maintenance plan to ensure the XGBoost model stays accurate, fair, and reliable over time.

Monitoring schedule

Weekly	Review **prediction logs** and flag outliers
Monthly	Compare **predicted vs actual OAR** outcomes
Quarterly	Full **model performance review** (AUC-ROC, F1)
Ad-hoc	Triggered review if **alert thresholds** are breached

Alert Thresholds

AUC-ROC score	`< 0.65 → retrain`
F1 score	`< 0.50 → investigate`
DREI positive rate	`shift > ±15%`
Actual vs predicted OAR	`divergence > 10%`

Retraining process

1. **Collect & validate new data**
   No missing values in key columns
2. **Re-run stage 2 & 3 notebooks**
   Feature engineering, SMOTE, hyperparameter tuning
3. **Evaluate vs current model**
   Must outperform current AUC-ROC (0.71) to replace
4. **Save & deploy new model**
   Push .joblib to GitHub → auto-deploys on Streamlit
5. **Validate dashboard output**
   Run sample predictions and verify results

The cost of guessing vs the value of predicting

Simulated on 2,000 recruitment records — comparing the full pipeline against only pursuing candidates predicted as High OAR (n=736). The numbers speak for themselves.

Cost Saved

$6.9M

From $10.82M → $3.90M saved on non-converting candidates

Days Saved

61,315 days

From 94,125 → 32,810 total days across all hiring processes

Candidates Focused on

736

Out of 2,000 total — only 36.8% needed for same outcome

OAR Improvement

+0.05

From 0.66 → 0.71 crosses the 0.70 threshold

Recruiter capacity freed

-63.2%

From 2,000 to 736 candidates to manage — recruiters spend **less time on low-probability candidates** and more on building relationships with likely acceptors.

Faster time-to-fill

-2.48 days

Faster avg time-to-hire means **open roles are filled sooner** — reducing business disruption, lost productivity, and the hidden cost of vacancy.

Decision quality improved

Data-driven

HR moves from **gut-feel to model-backed decisions** — every candidate comes with a probability score, factor breakdown, and a recommended action.

Strategic recommendations for smarter hiring

Immediate actions

Deploy RePort to HR teams

Stop evaluating candidates without data. **Open RePort, enter candidate details, and use the prediction score** to prioritize who gets fast-tracked — starting with the next open role.

Align sourcing to High OAR channels

Recruiter and Job Portal consistently outperform Referral and LinkedIn on OAR. **Shift sourcing budget toward higher-converting channels by department** — use the heatmap from the analysis.

Run what-if before every offer

Use RePort's Scenario Comparison tool before extending any offer. **Compare two sourcing or cost scenarios and choose the one with the higher predicted OAR** — it takes under 30 seconds.

Reduce pipeline volume — focus on fit

More applicants do not lead to better OAR. **Tighten job descriptions, add pre-screening steps, and target quality over quantity** — especially for roles with high applicants-per-day ratios.

Track predicted vs actual OAR monthly

Start logging prediction outcomes immediately. **Compare model predictions against real acceptance decisions** each month — this is the foundation for knowing when to retrain.

Long-term strategy

Integrate real-time HR data feed

Eliminate manual CSV uploads by connecting RePort directly to your HR system. **Live data means live predictions** — no lag, no manual effort, no risk of stale inputs affecting decisions.

Expand training data beyond 5,000 records

The model was trained on a limited dataset. **As more recruitment data is collected, retrain quarterly** — more data means better generalization, higher AUC, and more reliable subgroup predictions.

Enrich candidate data collection

The model currently relies on process metrics (cost, time, source) rather than candidate quality signals. **Integrating interview scores, assessment results, and background data** will shift the model from predicting process efficiency to predicting candidate-role fit — a fundamentally stronger signal for offer acceptance.

Extend model coverage

The current project applies multiple supervised learning algorithms — LR, KNN, SVM, DT, RF, XGBoost, Gradient Boosting — with **hyperparameter tuning constrained to feasible ranges** due to resource and time limitations.

UnderCode

Rakamin Academy

Hire Smarter Not Harder

A data-driven approach to optimizing recruitment efficiency and predicting offer acceptance

GitHub Dataset Python Code Dashboard